

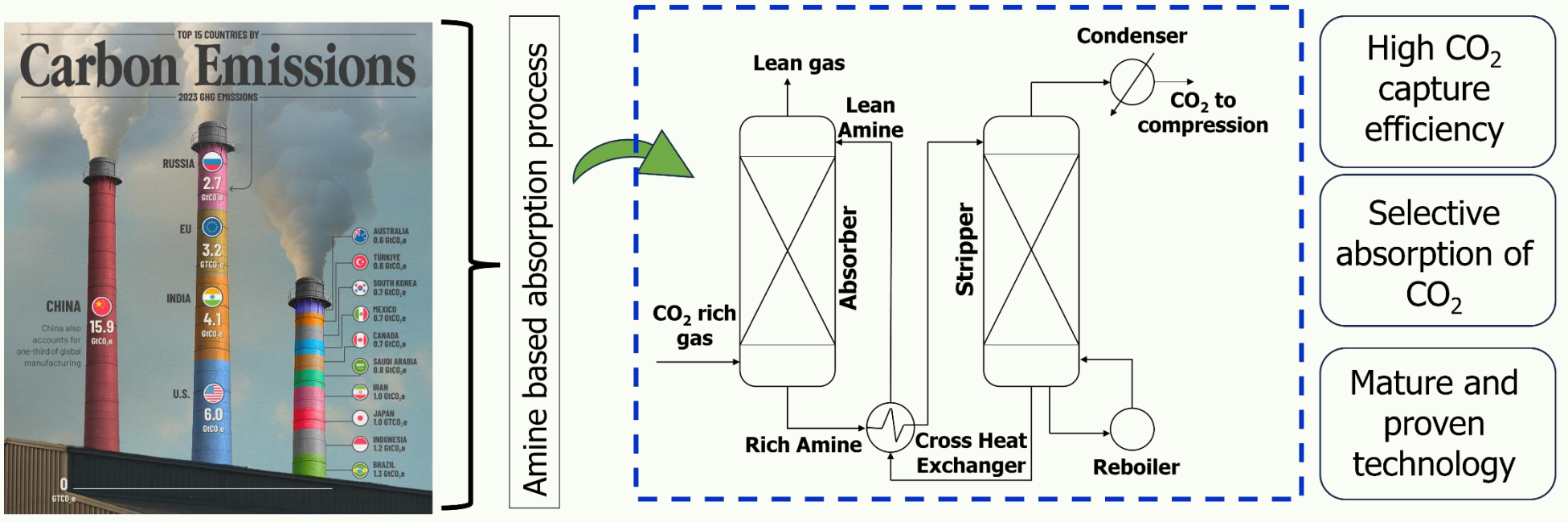
CFD Analysis of Reactive CO₂ Absorption in a Helical Static Mixer Structured Packing: A Comparison between Explicit Geometry and Porous Media Model

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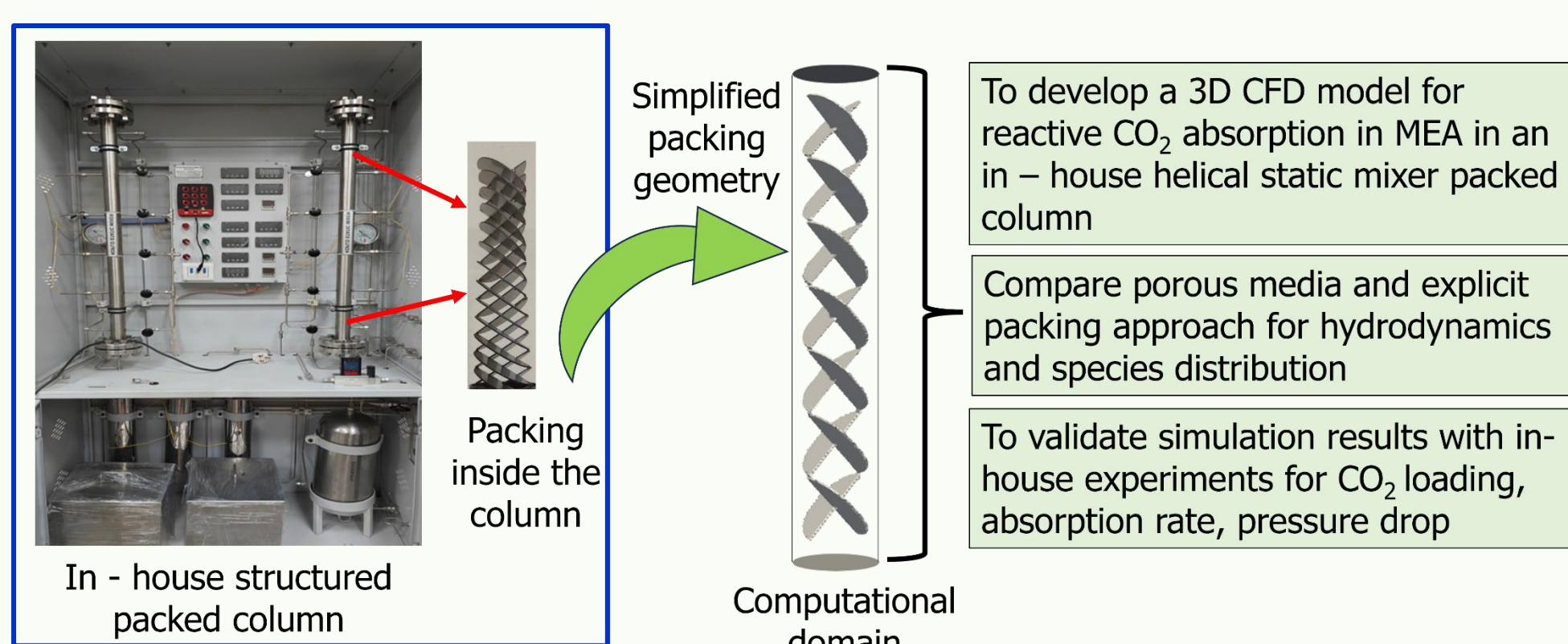
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01 INTRODUCTION



02 OBJECTIVES



03 METHODOLOGY

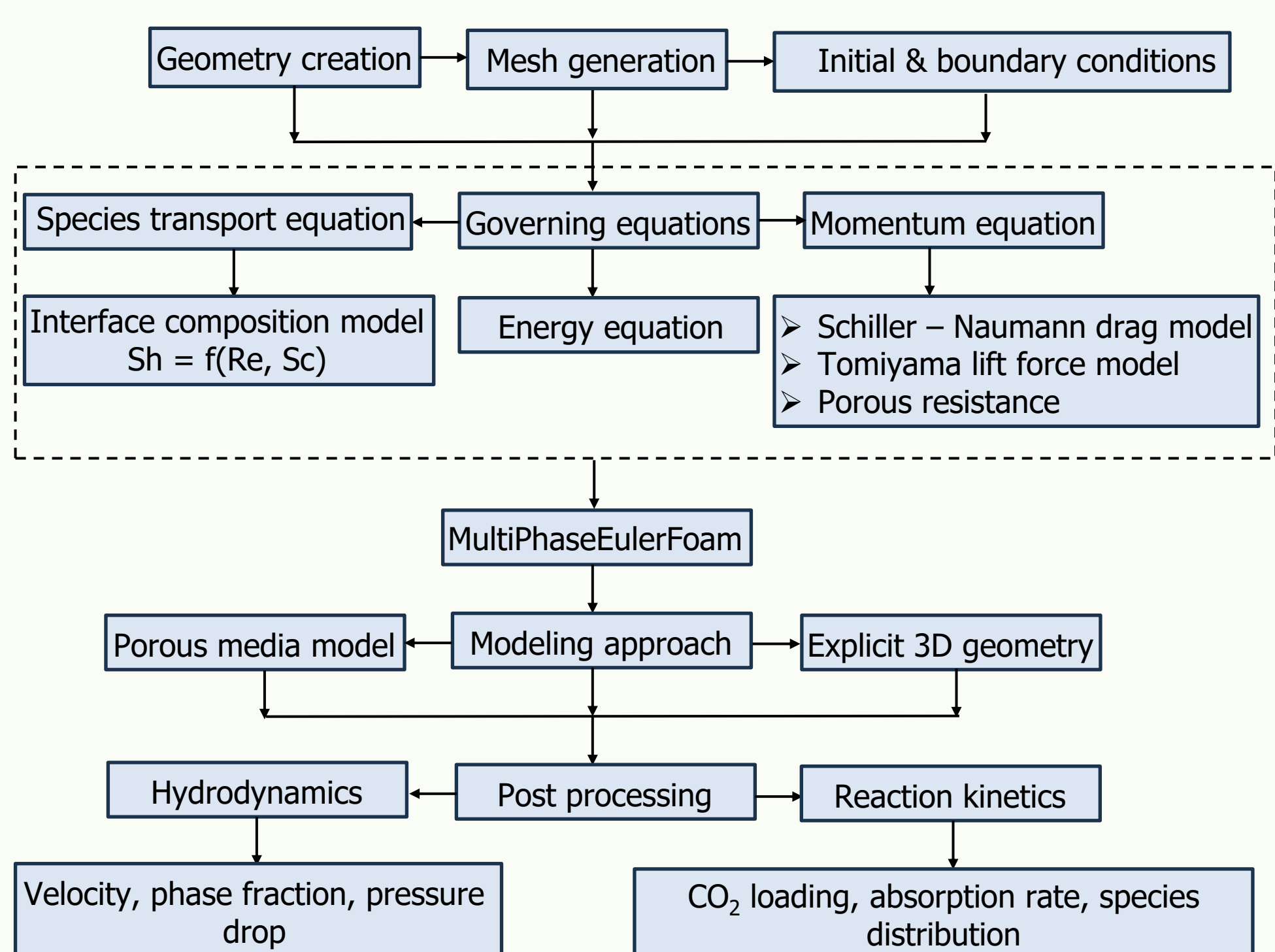


Figure 1. Methodology for CFD modeling of reactive CO₂ absorption in amines

Table 1. Initial and operating conditions

Parameter	Value
Height of column	28 cm
Diameter of the Column	5 cm
Diameter of packing	4.797 cm
Velocity of gas	0.021 m/s
Height of the liquid filled in the column	28 cm
Initial Temperature of gas	323 K
Initial temperature of the liquid	323 K
Initial mass fraction of MEA in liquid phase	0.2
Initial mass fraction of CO ₂ in gas phase	0.282
Initial mass fraction of H ₂ O in liquid phase	0.8
Initial mass fraction of air in gas phase	0.718

Table 2. Boundary conditions

Variable	Inlet	Outlet	Walls and Packing
Phase Fraction	fixedValue	zeroGradient	zeroGradient
Mass Fraction	fixedValue	inletOutlet	zeroGradient
Velocity	fixedValue	zeroGradient	noSlip
Temperature	fixedValue	zeroGradient	zeroGradient
Pressure	calculated	calculated	calculated

REFERENCES

- Zidouni, F. *et al.* (2015). *Chem. Eng. Sci.*, **137**, 476–486
- Chandra & Rabha (2025). *Energy Fuels*, **39**, 15822–15833

04 RESULTS

4.1. HYDRODYNAMICS

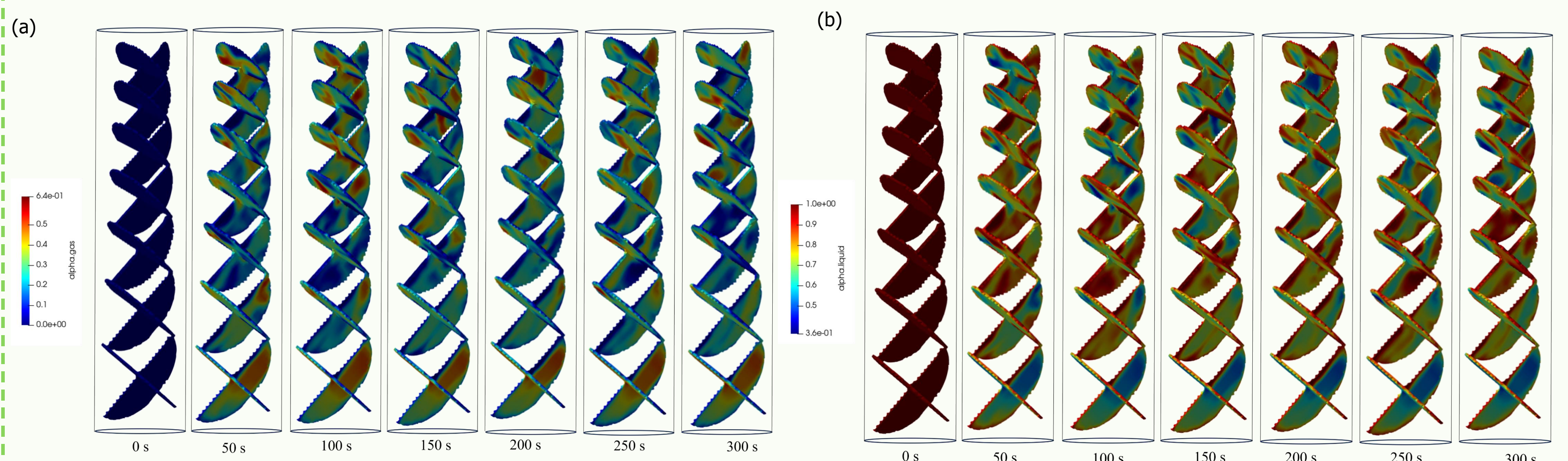


Figure 2. Snapshots of phase distribution of (a) gas (b) liquid

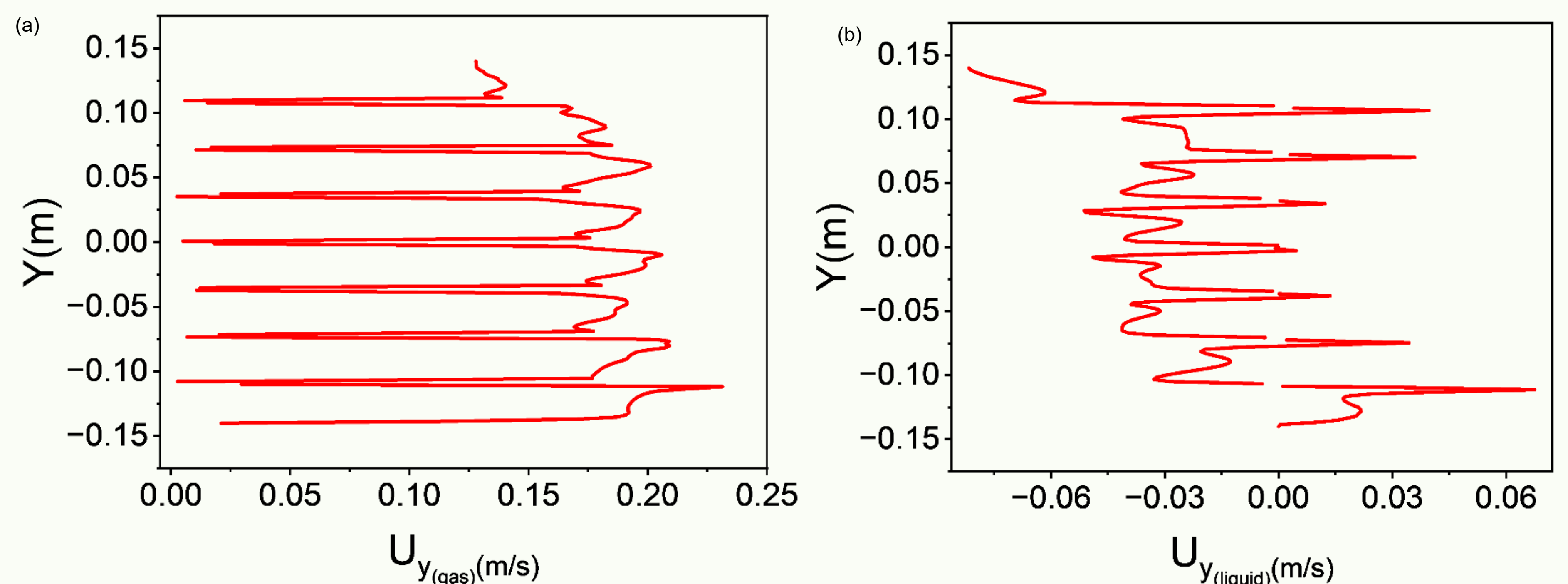


Figure 3. (a) Time averaged axial velocity of (a) gas (b) liquid

4.2. REACTION KINETICS

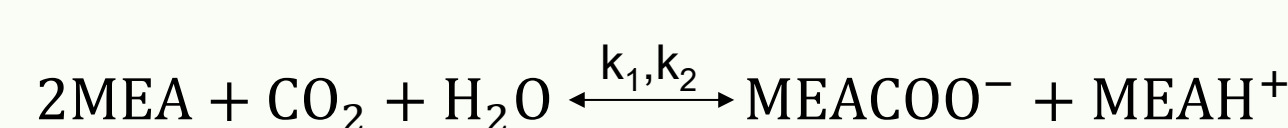


Table 3. Effect of gas flow rate on CO₂ loading

Flow rate of gas (LPM)	Experimental data (CO ₂ loading)	Simulation results (CO ₂ loading)
2	0.42	0.32
2.5	0.55	0.40
4.2	0.57	0.60

Table 4. Effect of gas flow rate on unit volume absorption rate

Flow rate of gas (LPM)	Experimental data (Unit volume absorption rate)	Simulation results (Unit volume absorption rate)
2	1.946	1.945
2.5	2.421	2.420
4.2	4.092	4.091

4.3. COMPARISON OF SPECIES DISTRIBUTION IN POROUS MEDIA APPROACH MODELLING AND EXPLICIT STRUCTURED PACKING MODELLING

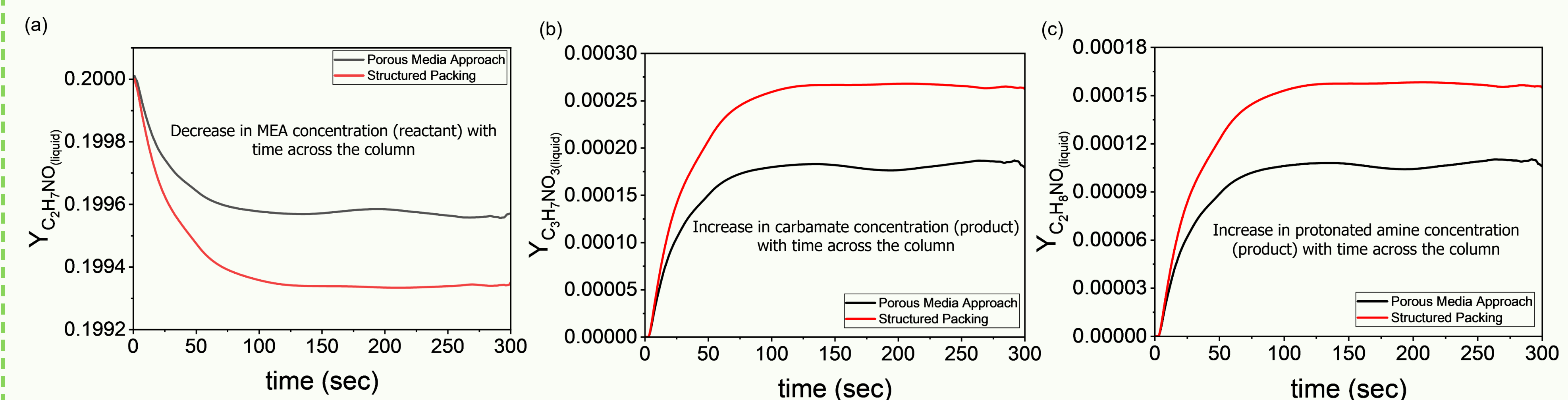


Figure 4. Distribution of (a) monoethanolamine (MEA) (b) carbamate (c) protonated amine

05 CONCLUSION

- Developed a 3D CFD Model for reactive absorption in MEA for both porous media and helical static mixer structured packing.
- Simulation results showed good agreement with in - house experimental data, showing the reliability of developed model.
- The explicit packing model captured the detailed flow and species distribution accurately than porous media approach, while porous media offered low computational cost.
- The developed CFD model enables the optimization of structured packing designs and the investigation of hydrodynamics in different packing geometries using ERT/ECT, provides insights into effect of hydrodynamics on CO₂ mass transfer performance.

ACKNOWLEDGEMENT